

# SOP-BL-03 — Usability Heuristic Evaluation

| Kode      | SOP-BL-03                                                                                                                   |
|-----------|-----------------------------------------------------------------------------------------------------------------------------|
| Pemilik   | Bulma (UI/UX Lead)                                                                                                          |
| Revisi    | 1.0 · <b>Berlaku</b> 2026-05-29 · <b>Reviu</b> per kuartal                                                                  |
| Referensi | Nielsen 10 Usability Heuristics, WCAG 2.x AA, ISO 9241, tools/heuristic-eval-checklist.md, tools/accessibility-checklist.md |
| COBIT     | BAI03 (Managed Solutions Build)                                                                                             |

## 1. Tujuan

Memastikan desain **usable & accessible** (bukan cuma cantik) — surface friction & pelanggaran heuristik/WCAG **sebelum** handoff, biar gak bocor ke user (anti "aesthetic > function").

## 2. Ruang Lingkup

- **Berlaku**: tiap flow/page baru atau yang diubah, sebelum handoff ke Killua (SOP-BL-04).
- **Tidak berlaku**: eksplorasi internal yang belum jadi flow utuh.

## 3. Definisi & Istilah

- **Heuristic evaluation** — audit cepat desain vs 10 aturan Nielsen.
- **WCAG AA** — kontras body  $\geq 4.5:1$ , large  $\geq 3:1$ ; focus state; motion-safe.
- **POUR** — prinsip WCAG: Perceivable, Operable, Understandable, Robust.
- **Severity** — bobot temuan: Blocker (a11y fail / flow rusak) / Major / Minor.

## 4. Referensi

Nielsen 10 (rule-of-thumb usability), WCAG 2.x (POUR), ISO 9241 part 11 (usability = effectiveness+efficiency+satisfaction), mandate F-2 boomer-proof.

## 5. Tanggung Jawab (RACI)

| Aktivitas            | R     | A     | C       | I |
|----------------------|-------|-------|---------|---|
| Heuristic eval       | Bulma | Bulma | —       | — |
| A11y check (WCAG AA) | Bulma | Bulma | @I (QA) | — |
| Fix temuan Blocker   | Bulma | Bulma | —       | — |

## 6. Prosedur

### 6.1 Heuristic pass (Nielsen 10)

- 6.1.1 Telusuri flow lewat **heuristic-eval-checklist**, cek tiap heuristik:
  - a. **H1 Status** — empty/loading/error state ada? user tau sistem lagi apa?
  - b. **H2 Match real world** — copy bahasa warung (F-2), ikon familiar, no jargon?
  - c. **H4 Consistency** — pakai token design-system, button identik lintas page?
  - d. **H5 Error prevention** — validasi visual, confirm destructive action?
  - e. **H6 Recognition>recall** — label jelas, gak ngandelin hafalan?
  - f. **H8 Minimalist** — 1 section 1 fokus, no noise?
  - g. **H9 Error recovery** — error state kasih tau cara benerin, bukan cuma merah?
- 6.1.2 Catat tiap pelanggaran + **severity**.

### 6.2 Accessibility pass (WCAG AA)

- 6.2.1 Jalankan **accessibility-checklist**: **kontras AA** (catat rasio), focus state kelihatan, target tap  $\geq 44\text{px}$ , motion-safe ( **prefers-reduced-motion** ), gak ngandelin warna doang, heading order benar.
- 6.2.2 **Decision point**: ada **a11y Blocker** (kontras fail / no focus)? → **wajib fix**, gak boleh handoff.

### 6.3 Verdict (exit)

- 6.3.1 **Lethal check**: 3-detik mobile message sampai? ibu-warung ngerti CTA? keep-3-element test?
- 6.3.2 Semua Blocker fixed + Major ditangani → **lolos eval**, lanjut handoff (SOP-BL-04). Ada Blocker → loop fix.

## 7. Pengecualian

- **Trade-off a11y vs estetika**: kalau ngorbanin AA demi estetika → **escalate Tata**  (gak boleh diputus sendiri); default tetap AA.

## 8. Rekaman & Retensi

| Rekaman                    | Lokasi                                  | Retensi  |
|----------------------------|-----------------------------------------|----------|
| Heuristic eval + severity  | Log.md + checklist terisi               | Permanen |
| A11y check (rasio kontras) | accessibility-checklist terisi + Log.md | Permanen |

## 9. Indikator Kinerja (KPI)

| KPI                       | Rumus                                          | Target      |
|---------------------------|------------------------------------------------|-------------|
| Heuristic violation lolos | # violation ketemu user/QA setelah handoff     | $\approx 0$ |
| Kontras AA compliance     | # pasangan teks lolos AA $\div$ total          | 100%        |
| State coverage            | # flow dengan empty/error/loading $\div$ total | 100%        |

## 10. Riwayat Revisi

| Rev | Tanggal    | Perubahan      |
|-----|------------|----------------|
| 1.0 | 2026-05-29 | Terbit pertama |